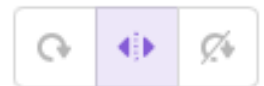


SCRATCH Pac-Man 1D



Play one-dimensional Pac-Man in Scratch.

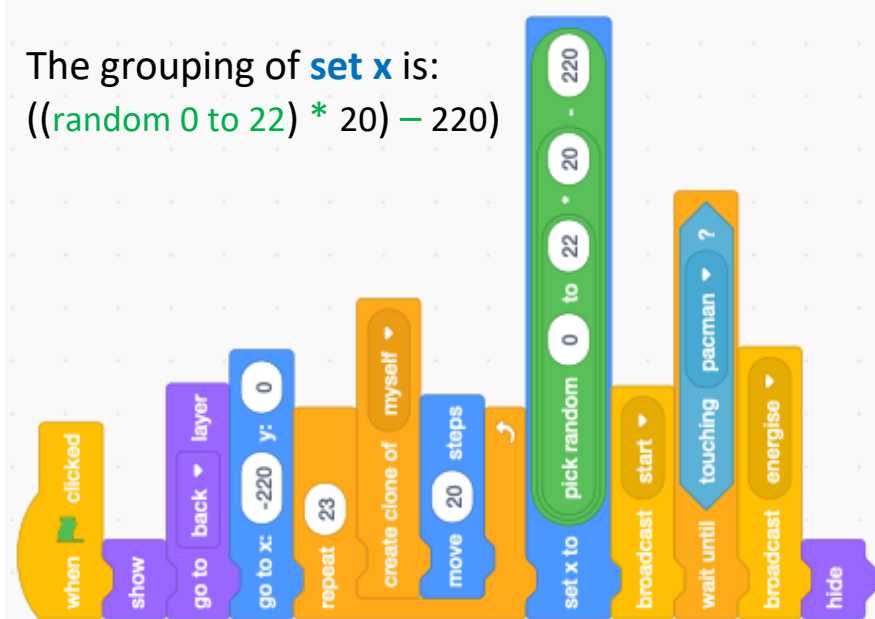
- 1) Register/login at <https://scratch.mit.edu>
- 2) Download images and create sprites from:
<https://codeclub67.github.io/images/pacman.gif>
<https://codeclub67.github.io/images/ghost.gif>
<https://codeclub67.github.io/images/pellet.gif>
- 3) Double the size of Pac-Man and the Ghost by setting their **size** to 200%.
- 4) Set the **direction** of Pac-Man and the ghost to mirror left and right, rather than rotate.
- 5) Add a dark background (e.g. Stars).



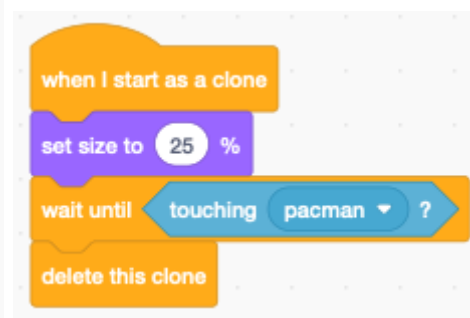
Pac-Man 1D has just a single row of 23 pellets to eat. These small (25% size) pellets are clones.

- 6) Add the code (below) to the **pellet**. The original pellet is

The grouping of **set x** is:
 $((\text{random } 0 \text{ to } 22) * 20) - 220$



the energiser
which is placed
randomly.

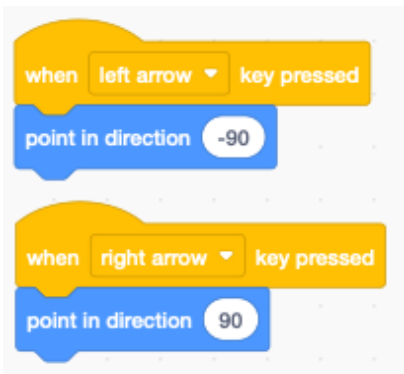
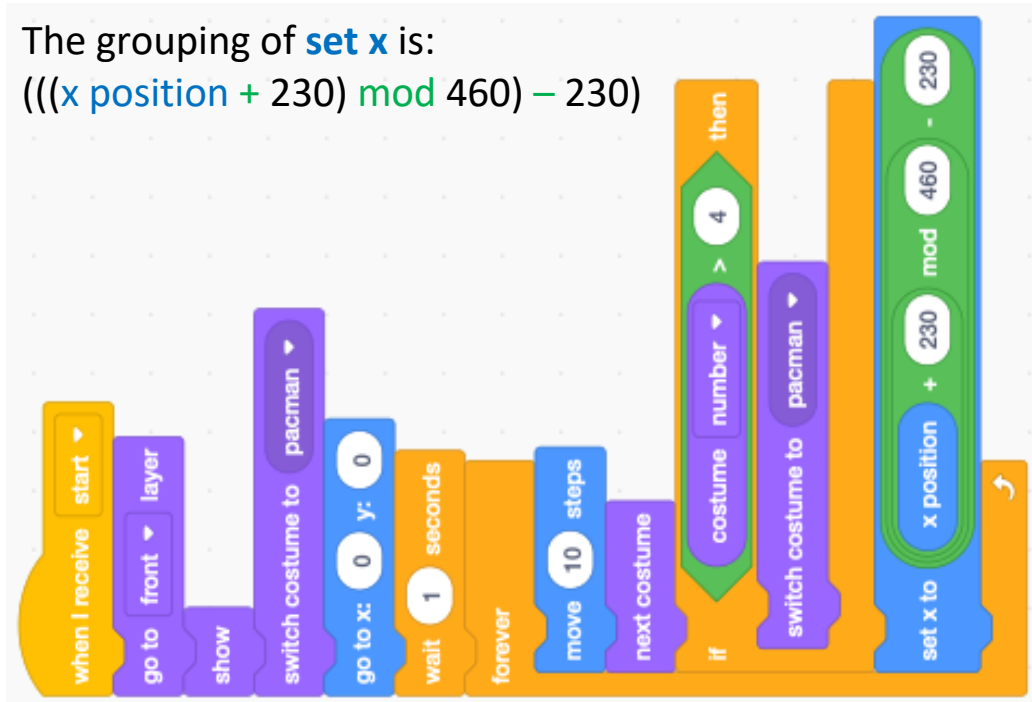


Cycle through Pac-Man's first four costumes to animate it.

7) Add the code below to **Pac-Man**. The **set x** block creates a wrap-around (1D) world for Pac-Man. If it goes off one edge it reappears at the other edge. The screen width is 460 and **mod** gives the remainder after division.

The grouping of **set x** is:

$((x \text{ position} + 230) \bmod 460) - 230$



8) Guide Pac-Man with **left/right keys** (see code left) and point it in the corresponding **direction**.

9) When Pac-Man loses and it's "**game over**", play

out the rest of

its costumes, up to costume 20,

before moving Pac-Man off-

screen (see code right).



When Pac-Man is energised it has two seconds to catch the ghost before it goes back to chasing Pac-Man. The energiser has a time limit of 2 seconds into the future using the **timer**. When the **timer** is bigger (later) than the **time limit** the normal ghost costume is displayed, and it heads towards Pac-Man. If it touches Pac-Man it's "game over."

However, if the time limit hasn't expired and the ghost touches Pac-Man, then the ghost 'dies' (costume 6 "eyes").

10) Create a new variable **time limit** and add the code below to the **ghost**. The ghost has a random starting position and it loops until the ghost 'dies'. The small loop at the end animates the 'death' of the ghost.

The grouping of **set x** is: $((x \text{ position} + 230) \bmod 460) - 230$

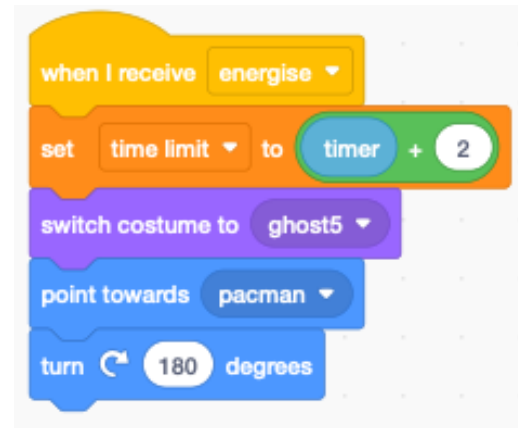


The set x block does the same wrap-around as in Pac-Man and the grouping of the blocks is the same.

The ghost reacts to the energiser by changing to the “frightened” costume and making a 180° turn away from Pac-Man.

11) Add the code (right) to the **ghost**.

Try duplicating the ghost and selecting a different character (Inky, Blinky, Pinky or Clyde), how hard is it to play?



Save your code with a good name. **File > Save now**

Pac-Man graphics sourced from:

<https://www.sprites-resource.com/arcade/pacman>